Drawing Direction Vectors in Cubic Cells

Original statement · Problems T2_CrystStruct.pdf, p. 3

> *12. Draw direction vectors in unit cells for the following cubic directions: a)* $[1\,1\,2]$ *b)* $[\bar{3}\,3\,1]$ *c)* $[2\,1\,2]$ *d)* $[\bar{1}\,0\,1]$ *e)* $[3\,2\,1]$ *f)* $[1\,2\,2]$ *g)* $[\bar{1}\,2\,3]$ *h)* $[0\,\bar{2}\,1]$ *i)* $[2\,\bar{3}\,3]$ *j)* $[1\,2\,\bar{1}]$ *k)* $[2\,2\,3]$ *l)* $[\bar{1}\,0\,3]$

PHASE 1: HYPOTHESIS & PARAMETERS

Geometric Context:

Twelve crystallographic vectors are drawn inside a standard cubic unit cell of normalised dimensions $1 \times 1 \times 1$.

Normalisation and Origin Translation Rules [Session 3 Slides 37-40]:

1. The components of the direction vector must be reduced by dividing by the component of largest absolute value ($u_{\max}$) so that the vector fits strictly inside the volume of a single unit cell:

$$r = \left(\frac{u}{u_{\max}}, \frac{v}{u_{\max}}, \frac{w}{u_{\max}} \right)$$

2. If any component is negative (overbar $\bar{u}$), the **origin of the vector must be translated** to the position $+1$ along that axis so that the vector points towards the interior of the cell.

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

Citing the systematic procedure [Session 3 Slide 40]:

1. Identify whether there are negative indices and fix the local origin:
2. $\bar{u} \implies$ origin at $x = 1$.
3. $\bar{v} \implies$ origin at $y = 1$.
4. $\bar{w} \implies$ origin at $z = 1$.
5. Draw the vector from the origin (x_0, y_0, z_0) to the end point $(x_0 + \Delta x, y_0 + \Delta y, z_0 + \Delta z)$.
6. Join both points with an oriented arrow.

PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP

PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION

1. **Parallelism:** If we multiply a vector by a positive scalar constant, the direction does not change: $[1\,1\,2]$ and $[2\,2\,4]$ represent exactly the **same crystallographic direction**.
2. **Equivalence of $\langle u\,v\,w \rangle$ Families:** Directions such as $[1\,2\,2]$ and $[2\,1\,2]$ are symmetric members of the $\langle 221 \rangle$ family in the cubic lattice, which implies that they have identical translational repeat length and the same physical properties of conductivity or elastic modulus.

Study notes written from the official course material (Problems T2_CrystStruct.pdf; Problems T2 defects.pdf).
Online version: subjects/aerospace-materials-1/problemas/topic-2-problems.html